

Iftekhar Ibne Masud Rafi

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ABOUT ME

I am a final-year Software Engineering student at Zhengzhou University, China, with a strong foundation in programming, software development, and computer science. My academic and project experience has developed my interests in **Artificial Intelligence, Machine Learning, Computer Vision, Natural Language Processing, and Web Development**. I am particularly interested in exploring emerging technologies through advanced study and research, with the goal of applying computing to solve real-world challenges.
I am motivated by continuous learning, problem-solving, and the opportunity to turn ideas into meaningful technological solutions.

EDUCATION AND TRAINING

Bachelor's

Zhengzhou University [10/09/2013 – Current]

Address: Address: 100 Kexue Blvd, Zhongyuan District, Zhengzhou, Henan, 450001 Zhengzhou (China) | **Website:** https://english.zzu.edu.cn/

Higher Secondary Certificate (HSC)

Savar Government College [01/06/2019 – 25/12/2021]

Address: 41 Thana Road, Savar, Dhaka, Bangladesh, 1340 Dhaka (Bangladesh) | **Website:** https://www.savargc.gov.bd/en

Secondary School Certificate (SSC)

Chapain New Model High School [01/01/2017 – 06/05/2019]

Address: Chapain, Savar, Dhaka, 1340 Dhaka (Bangladesh)

PROJECTS

Automatic Number Plate Recognition (ANPR) system Engineered an end-to-end ANPR system using YOLO, OpenCV, and EasyOCR, with FastAPI, MongoDB Atlas, and React for backend, data storage, and monitoring.

Medha AI: AI-Powered Intelligent Assistant
Developed a full-stack AI assistant using React, TypeScript, FastAPI, and Groq Llama 3.3 70B, with context-aware chat, session management, persistent history, Markdown/code rendering, and responsive UI.

SkyCast: Weather Forecasting Web Application
Developed a responsive weather forecasting app using React, TypeScript, and Open-Meteo APIs, with real-time forecasts, location detection, Redux Toolkit, TanStack Query, and Tailwind CSS.

EmotionVision: Real-Time Facial Emotion Recognition System
Developed a real-time facial emotion recognition app using Python, OpenCV, PySide6, and deep learning, featuring face detection, emotion classification, confidence scoring, and a responsive GUI.

Handwritten Character Recognition Developed a CNN-based MNIST digit classification system, achieving 98.94% test accuracy with model comparison, error analysis, and inference benchmarking. Gained hands-on experience in deep learning, computer vision, and model evaluation.

Disease Prediction from Medical Data Machine learning project for predicting heart disease from clinical data(UCI Heart Disease dataset). It compares three classifiers and applies threshold optimization to prioritize reducing false negatives.

Airline Management System
Developed a full-stack airline management system using PySide6, React, FastAPI, and PostgreSQL, with flight search, booking, payments, e-tickets, refunds, JWT authentication, and admin management.

MiniRedis: Redis-Inspired In-Memory Database
Built a lightweight Redis-inspired key-value database in Java with a custom TCP server, RESP protocol, multiple data structures, TTL management, and file-based persistence.

Language Model Evaluation Harness An NLP and Large Language Model (LLM) evaluation project designed to measure whether changes to prompts improve or degrade language-model behaviour.

HarmonyCare – Smart Elderly Safety Network | Team Project Collaborated with a team to develop an Android elderly safety application as part of the Mobile Programming course, using Java, Android SDK, XML, and MVVM. Contributed to UI development, Room/SQLite database management, SOS workflows, GPS/AMap integration, notifications, and accessibility features.

Interactive Personal Portfolio Developed a responsive personal portfolio using React, Vite, Tailwind CSS, Three.js, and Framer Motion, featuring interactive 3D elements, animations, project showcases, resume integration, and optimized performance.

On-Campus Stadium Booking System | SRS Course Team Project Designed the UI/UX of a mobile-first On-Campus Stadium Booking System using Figma as part of the Software Requirements Specification (SRS) course.

Multilingual Travel & Tourism Website
Developed a responsive travel platform using React, Vite, Tailwind CSS, and Framer Motion, featuring destinations, travel stories, galleries, tour packages, and English/Bengali multilingual support.

Student Management System | Object-Oriented Techniques Course Project
Developed a Python/Tkinter student management system with admin authentication, CRUD operations, MySQL integration, search, and CSV export using OOP and three-tier architecture.

Office Supply Management System | Object-Oriented Principles and Java Desktop app for managing office supplies and teacher requisitions. Course project for Object-Oriented Principles and Java (2025). Built with a DAO/Service/UI layered architecture and JDBC-backed MySQL persistence.

ACADEMIC RESEARCH(COURSE BASED)

Multi-Device Communication Network Using USART, Timer, and DMA | Embedded Systems

Conducted course-based research for my Embeded System Course on designing a multi-device embedded communication network using **USART, hardware timers, and DMA** for reliable, low-latency data exchange. Analyzed master–slave communication, time-division scheduling, frame protocols, error detection, and DMA-based CPU offloading through prototype testing.

HPC as the Foundation of Generative AI: Parallel Computing Architectures and Scalability Challenges for LLMs

Conducted course-based research on how **HPC, GPU clusters, parallel processing, and distributed computing** support the training and deployment of Generative AI and LLMs. Analyzed parallel computing architectures, scalability challenges, and future research directions in large-scale AI systems for my **High-Performance Computing** Course

Dry Bean Variety Classification Using Machine Learning Algorithms

Conducted course-based research for my Artificial Intelligence course on classifying **dry bean varieties** using Decision Tree, KNN, SVM, and Random Forest algorithms. Evaluated model performance using accuracy, precision, recall, F1-score, and confusion matrices, with **SVM achieving 92.21% accuracy**.

A Survey of Wi-Fi Technologies: Evolution, Performance, and Applications

Conducted course-based research for my Computer Network course on the evolution of **IEEE 802.11 Wi-Fi technologies from 802.11b to Wi-Fi 7**, focusing on performance and key technologies such as MIMO, OFDMA, beamforming, and MU-MIMO. Reviewed security, interference, network congestion, and emerging applications of Wi-Fi.

SKILLS

Programming Language

Python / Java / JavaScript / C / TypeScript / HTML / SQL

Frameworks & Libraries

OpenCV / NumPy / PySide6 / Matplotlib / Scikit-learn / Pandas / React / Node.js / Express.js / FastAPI / Tailwind CSS / Framer Motion / TanStack Query / Redux Toolkit

Databases & Data Technologies

PostgreSQL / MongoDB Atlas / MySQL / Firebase / SQLAlchemy / Database Design

Systems & Core Technologies

Object-Oriented Programming / Software Architecture / System Design / Design Patterns / Multithreading

Web & Software Development

Full-Stack Development / REST APIs / Responsive Web Development / Authentication & Authorization / JWT / Git / GitHub

Development Tools & IDEs

Visual Studio Code / PyCharm / Android Studio / ItelliJ IDEA

UI/UX & Design

UI/UX Design / Figma / Adobe Photoshop / Canva

Other Software

Microsoft Word / Excel / PowerPoint / Adobe Premiere Pro / Capcut

CERTIFICATIONS

[School of International Education, Zhengzhou University., 25/10/2025]
China International College Student Innovation Competition 2025 Our team's project "UrbanAI: AI-Driven Urban Planning Platform for Smart Cities" has been recognized with a Certificate of Honor at the China International College Students' Innovation Competition (2025), awarded by the School of International Education, Zhengzhou University.

Mode of learning: Project based

[LinkedIn Learning]
Artificial Intelligence Foundations: Machine Learning Gained foundational knowledge of Artificial Intelligence and Machine Learning concepts, including core algorithms, models, and practical applications.

Mode of learning: Online

[Oxford Home Study Centre, 26/12/2024]
Machine Learning Strengthened my understanding of core Machine Learning concepts, techniques, algorithms, and their practical applications.

Mode of learning: Online

[HP Foundation, 07/04/2026]
Professional Networking for Career Growth In this course I have learned about the importance of networking for career growth. I have explored practical networking strategies, gained insight into the role of personal branding, and used digital tools to develop the skills needed to grow professional network with confidence.

Mode of learning: Online

[Zhegzhou University(ZZU-ISA), 10/03/2024]
Second ZZU International Student Eassay Contest Participated in the 2nd ZZU International Student Essay Contest and won the second prize, demonstrating strong academic writing, critical thinking, and communication skills.

Mode of learning: Work based

[Zhegzhou University, 10/10/2024]
Volunteer Certificate This certificate was presented to me for participation and for my outstanding contribution in the new student orientation volunteer activity organized by School Of International Education of Zhengzhou University in September 2024.

Mode of learning: Work based

Desinger Aknowledgement Represented in the BPL Cricket Tournament organized by the School of International Education at Zhengzhou University for working as the Graphics Designer.

BPL Cricket Tournament Committee Represented in the BPL Cricket Tournament organized by the School of International Education at Zhengzhou University for honor of being a member of tournament organizing committee.

LANGUAGE SKILLS

Mother tongue(s): Bengali

Other language(s): English | Hindi | Urdu | Spanish | Chinese

HOBBIES AND INTERESTS

Travel & Cultural Exploration

Photography & Visual Storytelling

Reading Novels

Exploring New Technologies & Ideas

Football & Cricket

RECOMMENDATIONS

Name: Li Yafei | Professor | Vice Dean
School of Computer Science and Artificial Intelligence
Zhengzhou University
Zhengzhou, Henan, China
Email: ieyfli@zzu.edu.cn

Name: Xiaokang Wang | Professor | Assistant of Dean
School of Computer Science and Artificial Intelligence
Zhengzhou University
Email: kxwang@zzu.edu.cn